

CHP-4 Electrostatic Potential

General Physics 2

2026-09-23

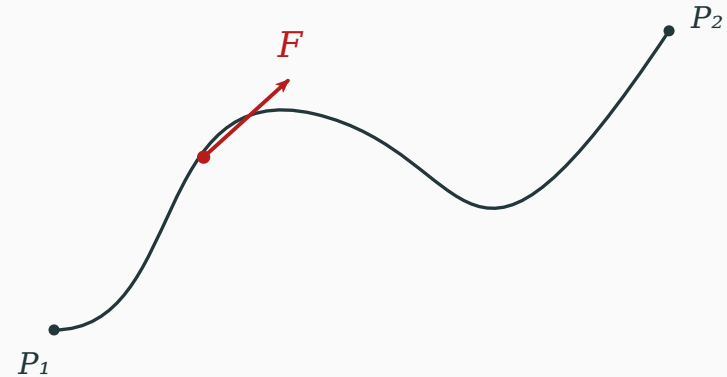
Work and Potential Energy

Work

Before defining electrostatic potential, let us first remember the concepts of **work** and **conservative forces**.

The work done by a force $\vec{F}$ on a particle as it moves along a path from P_1 to P_2 is

$$W = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{r}$$



$$W = \int F \cdot dr$$

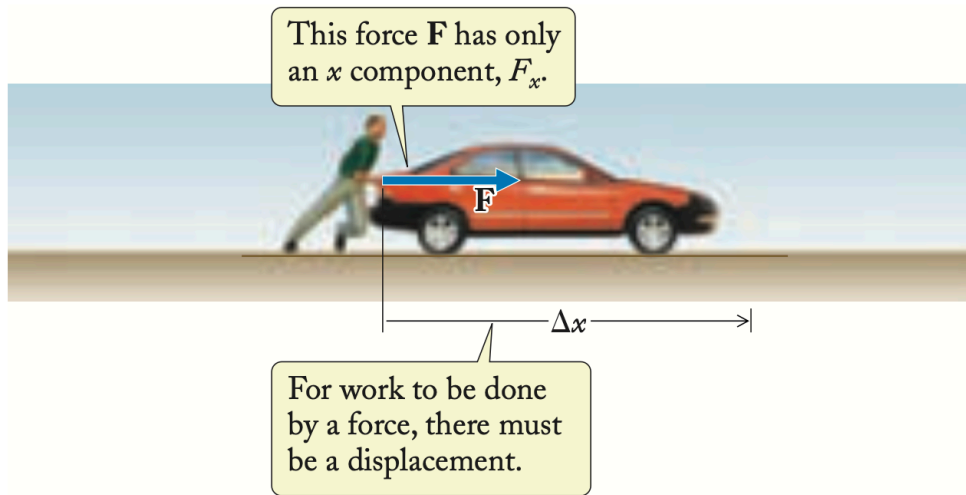


FIGURE 7.1 You do work while pushing an automobile along a road with a horizontal force $\mathbf{F}$.

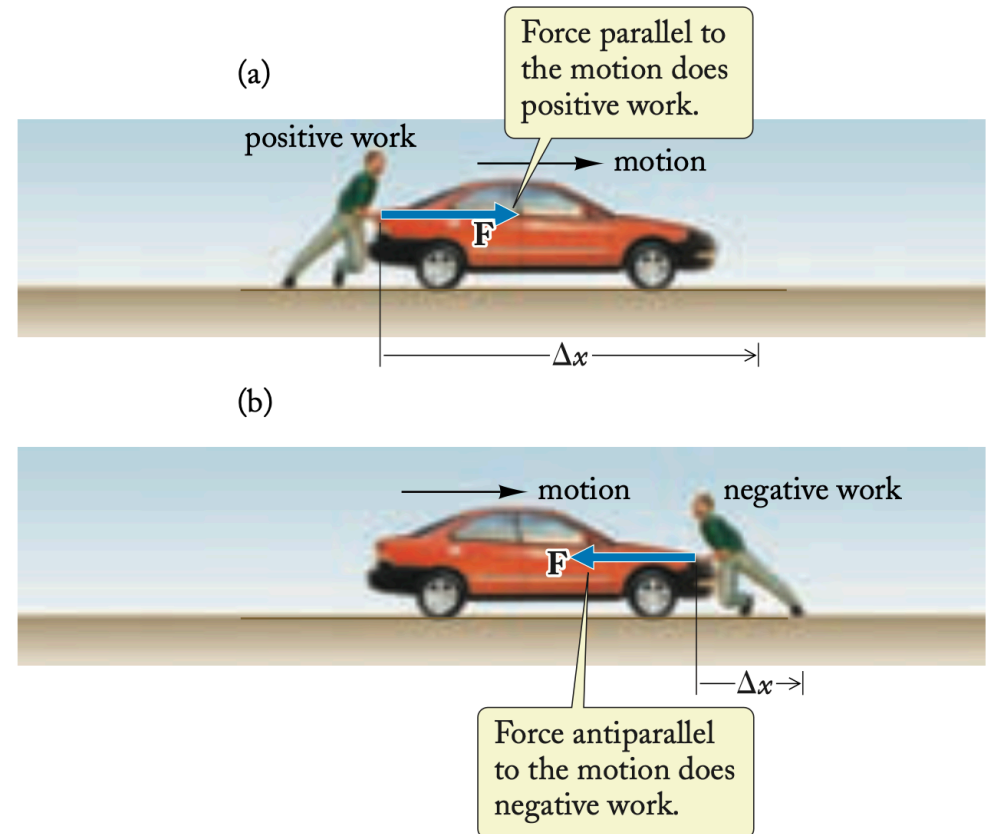
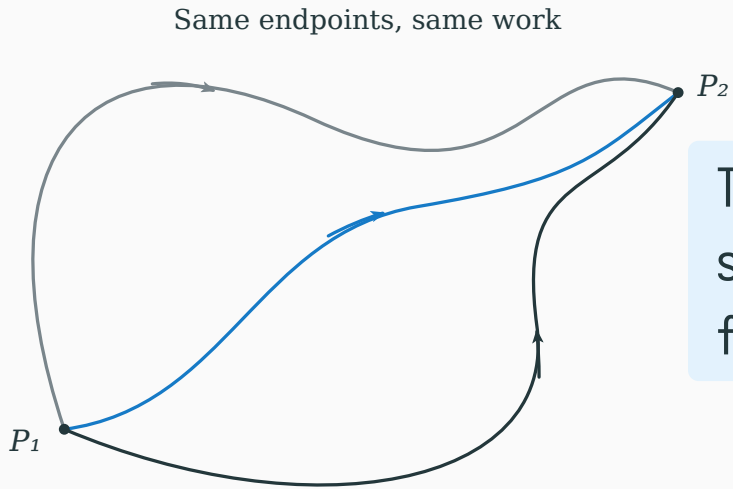


FIGURE 7.2 (a) The work you do on the automobile is positive if you push in the direction of motion. (b) The work you do on the automobile is negative if you push in the direction opposite to the motion.

Conservative Forces

Consider a particle moving from P_1 to P_2 along some path. Suppose the force $\vec{F}$ acting on the particle is a function of position only.

The force is **conservative** if its work depends only on the positions of P_1 and P_2 , not on the shape of the path between them.



The work done along different paths between the same endpoints is the same for a conservative force.

James Prescott Joule



James Prescott Joule (jool), 1818–1889, English physicist. He established experimentally that heat is a form of mechanical energy, and he made the first direct measurement of the mechanical equivalent of heat. By a series of meticulous mechanical, thermal, and electrical experiments, Joule provided empirical proof of the general law of conservation of energy.

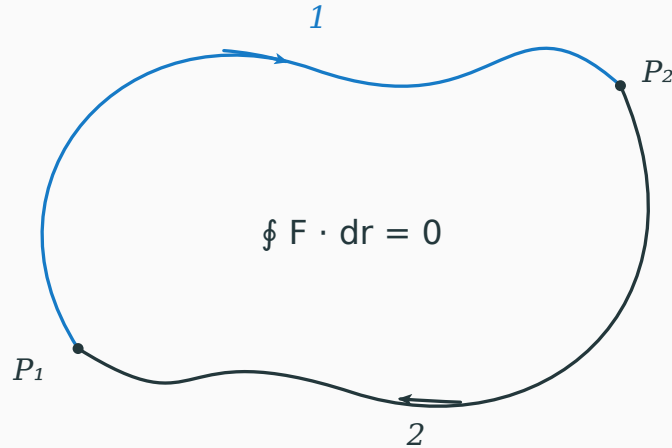
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Work Along a Closed Path

We can also express the conservative nature of a force in another way. Consider the following paths:



For a conservative force,

$$\int_{P_1}^{P_2} \vec{F} \cdot d\vec{r} + \int_{P_2}^{P_1} \vec{F} \cdot d\vec{r} = 0.$$

The first integral follows path 1 and the second follows path 2.

$$\oint \vec{F} \cdot d\vec{r} = 0$$

Potential Energy of a Conservative Force

The work done by a conservative force around a closed path is zero.

- Gravitational and Coulomb forces are conservative.
- Friction is an example of a nonconservative force.

Whenever a conservative force acts on a particle, we can define its potential energy at a point P by

$$U(P) = - \int_{P_0}^P \vec{F} \cdot d\vec{r} + U(P_0).$$

Joseph Louis Lagrange



Joseph Louis, Comte Lagrange
(lagranj) 1736–1813, French mathematician and theoretical astronomer, director of the Berlin Academy and professor at the École Polytechnique, Paris. He was the most eminent mathematician of the eighteenth century, barring only the great Leonhard Euler. In the elegant mathematical treatise *Analytical Mechanics* — which is probably the only book on physics completely devoid of diagrams — Lagrange formulated Newtonian mechanics in the language of advanced calculus and introduced the general definition of the potential-energy function. Lagrange is also known for his calculations of the motion of planets and for his influential role in securing the adoption of the metric system of units.

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His **Analytical Mechanics** formulated Newtonian mechanics in the language of advanced calculus and introduced a general definition of the potential-energy function.

He also studied planetary motion and helped secure the adoption of the metric system.

The Reference Point

P_0 is a reference point that we choose, and $U(P_0)$ is the potential energy we assign to it. For example, we may choose $U(P_0) = 0$.

The potential energies at two points are then

$$U(P_1) = - \int_{P_0}^{P_1} \vec{F} \cdot d\vec{r} + U(P_0),$$

$$U(P_2) = - \int_{P_0}^{P_2} \vec{F} \cdot d\vec{r} + U(P_0).$$

Potential-Energy Difference

Subtracting the two expressions gives

$$\begin{aligned}U(P_2) - U(P_1) &= - \int_{P_0}^{P_2} \vec{F} \cdot d\vec{r} + \int_{P_0}^{P_1} \vec{F} \cdot d\vec{r} \\&= - \int_{P_0}^{P_2} \vec{F} \cdot d\vec{r} - \int_{P_1}^{P_0} \vec{F} \cdot d\vec{r}.\end{aligned}$$

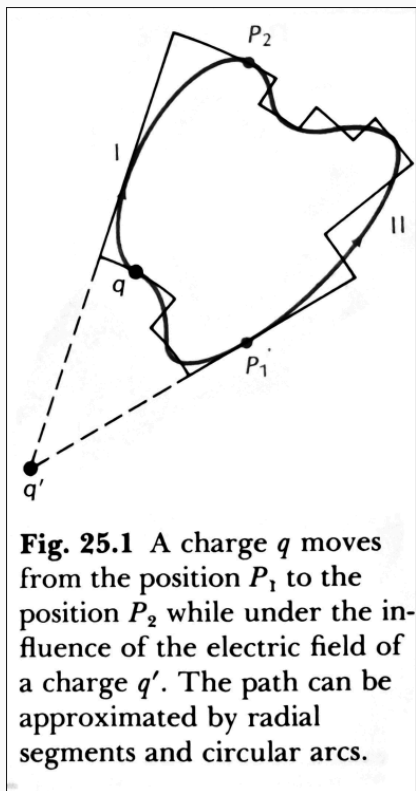
$$U(P_2) - U(P_1) = - \int_{P_1}^{P_2} \vec{F} \cdot d\vec{r}$$

The reference value $U(P_0)$ cancels. The physically significant quantity is the **change in potential energy**.

Electrostatic Potential

The Coulomb Force Is Conservative

Now apply the definition of potential energy to the electric force exerted by a fixed point charge q' on a charge q .



A charge q moves from P_1 to P_2 in the electric field of q' .

- Along circular arcs, $\vec{F} \cdot d\vec{r} = 0$: the force is perpendicular to the displacement.
- Along radial segments, the work is nonzero.

Only the radial displacement contributes to the work.

Electrostatic Potential

Electrostatic potential is the potential energy per unit charge:

$$V(P) = \frac{U(P)}{q}.$$

Since $\vec{F} = q\vec{E}$,

$$V(P) = - \int_{P_0}^P \frac{\vec{F}}{q} \cdot d\vec{l} + \frac{U(P_0)}{q}.$$

$$V(P) = - \int_{P_0}^P \vec{E} \cdot d\vec{l} + V(P_0)$$

Potential Difference

$$V(P_2) - V(P_1) = - \int_{P_1}^{P_2} \vec{E} \cdot d\vec{l}$$

The potential difference is the work per unit charge that an external agent must do against the electric force to move a charge from P_1 to P_2 , without changing its kinetic energy.

The unit of electrostatic potential is the **volt**:

$$1 \text{ volt} = 1 \text{ V} = 1 \text{ joule/coulomb} = 1 \text{ J/C.}$$

Alessandro Volta



Alessandro, Conte Volta, 1745–1827, Italian physicist, professor at Pavia. Volta established that the “animal electricity” observed by Luigi Galvani, 1737–1798, in experiments with frog muscle tissue placed in contact with dissimilar metals, was not due to any exceptional property of animal tissues but was also generated whenever any wet body was sandwiched between dissimilar metals. This led him to develop the first “voltaic pile,” or battery, consisting of a large stack of moist disks of cardboard (electrolyte) sandwiched between disks of metal (electrodes).

Alessandro, Conte Volta (1745–1827), Italian physicist and professor at Pavia.

He showed that the “animal electricity” observed by Luigi Galvani was not a special property of animal tissue: wet material between dissimilar metals could generate it.

This led him to develop the first **voltaic pile**, a battery made of moist cardboard disks between metal disks.

Units of Electric Field

Remember that $\vec{F} = q\vec{E}$, so the electric field has units of newtons per coulomb.

In terms of volts,

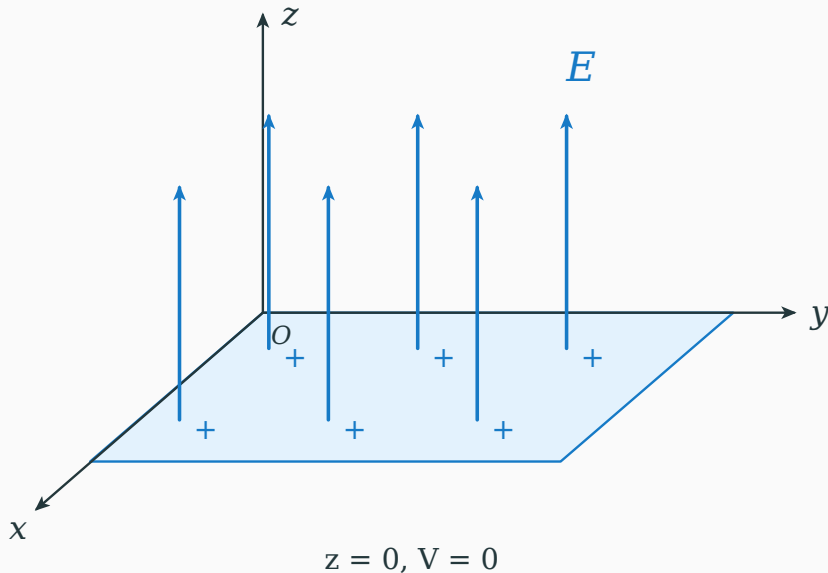
$$1 \text{ N/C} = 1 \frac{\text{N} \cdot \text{m}}{\text{C} \cdot \text{m}} = 1 \frac{\text{J}}{\text{C}} \frac{1}{\text{m}} = 1 \text{ V/m}.$$

$$1 \text{ N/C} = 1 \text{ V/m}$$

Example: A Large Conducting Surface

Problem: A very large, flat conducting surface carries a uniform surface charge density and generates a constant electric field $\vec{E}$.

What is the electrostatic potential at a distance z above the surface? Assume the potential is zero at the surface.



The electric field is vertical and independent of distance:

$$\vec{E} = E\hat{z}, \quad V(0) = 0.$$

Potential Above the Surface

Taking P_0 on the surface,

$$\begin{aligned} V(P) = V(z) &= - \int_{P_0}^P \vec{E} \cdot d\vec{l} \\ &= - \int E_x dx - \int E_y dy - \int E_z dz \\ &= - \int_0^z E dz' = -Ez. \end{aligned}$$

$$V(z) = -Ez$$

Conductors and Ground

The electric field inside a conducting body in electrostatic equilibrium is zero.

All points within the material and on the surface of a connected, homogeneous conductor are at the same potential.

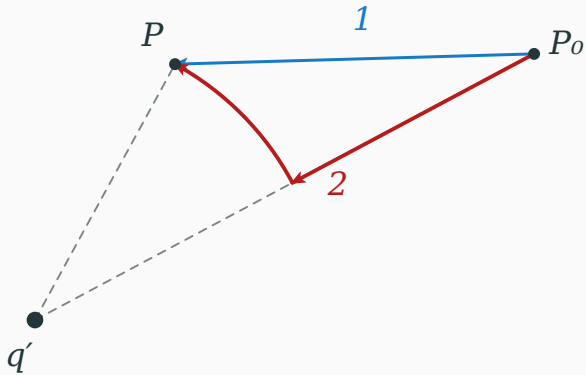
In experiments, it is often convenient to adopt the convention that the Earth's potential is zero, $V = 0$. The Earth then serves as **electrical ground**.

Calculation of Electrostatic Potential

Potential of a Point Charge

For a fixed point charge q' , start from

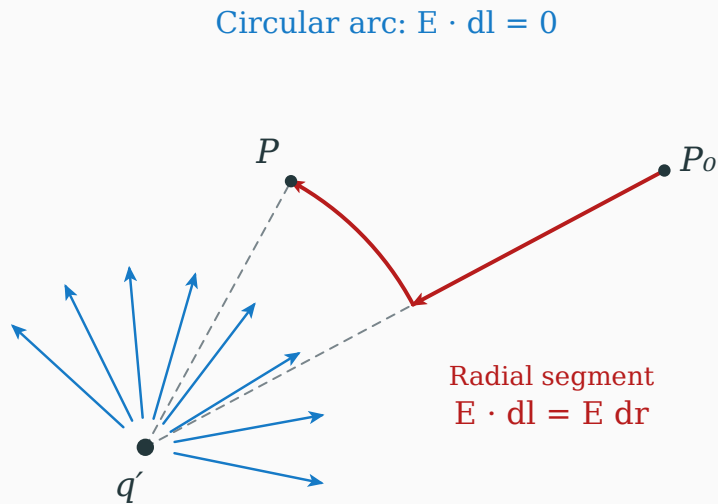
$$V(P) = - \int_{P_0}^P \vec{E} \cdot d\vec{l} + V(P_0).$$



The integral is independent of the path between P_0 and P . We may choose path 1 or path 2.

A Convenient Integration Path

Path 1 is harder to integrate. Path 2 consists of a radial segment and a circular arc.



The circular arc makes no contribution because $\vec{E} \cdot d\vec{l} = 0$.

Along the radial segment,

$$\vec{E} \cdot d\vec{l} = \left(\frac{1}{4\pi\epsilon_0} \frac{q'}{r^2} \hat{r} \right) \cdot (dr \hat{r}) = \frac{q'}{4\pi\epsilon_0 r^2} dr.$$

Point-Charge Potential

Let P be at radius r and P_0 at radius r_0 .

$$\begin{aligned} V(r) &= - \int_{r_0}^r \frac{q'}{4\pi\epsilon_0 r'^2} dr' + V(r_0) \\ &= \frac{q'}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{r_0} \right) + V(r_0). \end{aligned}$$

Choose the reference $r_0 \rightarrow \infty$ with $V(\infty) = 0$:

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q'}{r}$$

The potential energy of a charge q in this field is

$$U(r) = qV(r) = \frac{1}{4\pi\epsilon_0} \frac{qq'}{r}.$$

Example: An Electron Near a Proton

Problem: An electron is initially at rest at a very large distance from a proton. Under electric attraction, it falls toward the proton, which remains approximately at rest.

What is the electron's speed when it has fallen to within 0.53×10^{-10} m of the proton?

The electrostatic potential generated by the proton at this distance is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q'}{r} = \frac{1}{4\pi\epsilon_0} \frac{1.6 \times 10^{-19} \text{ C}}{0.53 \times 10^{-10} \text{ m}} \approx 27 \text{ V}.$$

Electronvolts

For the electron, $q = -e = -1.6 \times 10^{-19} \text{ C}$. Its potential energy is

$$U = qV = -e \times 27 \text{ V} \approx -4.3 \times 10^{-18} \text{ J}.$$

For atomic physics it is often more convenient to use electronvolts:

$$U \approx -27 \text{ eV}$$

$$1 \text{ eV} = (1.6 \times 10^{-19} \text{ C})(1 \text{ V}) = 1.6 \times 10^{-19} \text{ J}.$$

Potential from a Group of Charges

Previously, we used the electric field to find the potential of a charge distribution. We can also divide the distribution into small pieces dq , each contributing

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

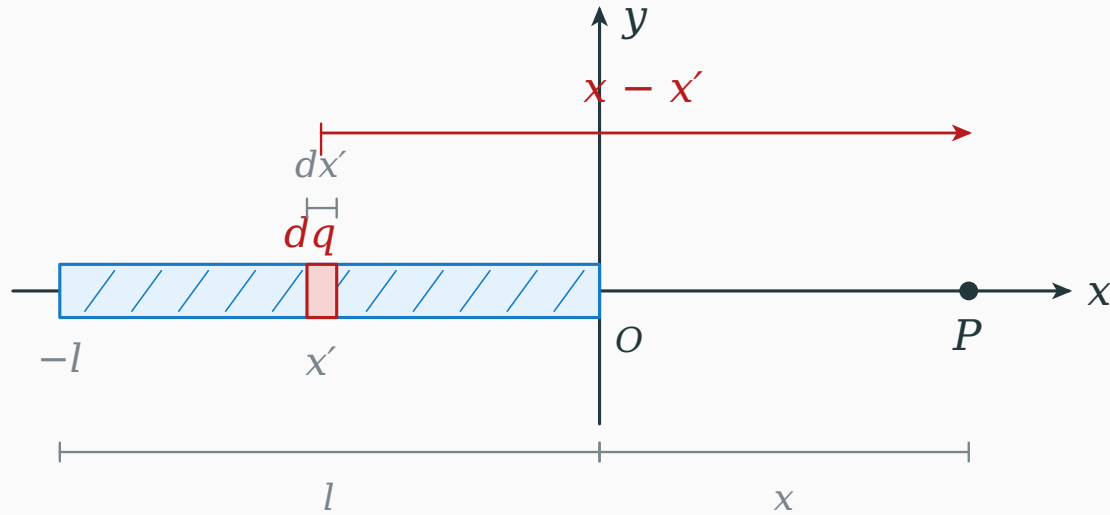
to the potential at the observation point, where r is its distance from dq .

Potential is a scalar. We add contributions without resolving them into directional components.

Be careful with the signs of the charges.

Example: A Uniformly Charged Rod

Problem: A rod of length l has charge Q uniformly distributed along its length. Find the electrostatic potential at a distance x from the end of the rod, on its axis.



$$dq = (Q/l) dx'$$

Take the rod along $-l \leq x' \leq 0$ and the observation point at $x > 0$.

Each element is a distance $x - x'$ from P . Note that x' is negative.

Potential on the Rod's Axis

Each element contributes

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{x - x'} = \frac{1}{4\pi\epsilon_0} \frac{\left(\frac{Q}{l}\right)dx'}{x - x'}.$$

Integrate over the length of the rod, taking $V(\infty) = 0$:

$$\begin{aligned} V(x) &= \frac{1}{4\pi\epsilon_0} \frac{Q}{l} \int_{-l}^0 \frac{dx'}{x - x'} \\ &= \frac{1}{4\pi\epsilon_0} \frac{Q}{l} [-\ln(x - x')]_{-l}^0. \end{aligned}$$

$$V(x) = \frac{1}{4\pi\epsilon_0} \frac{Q}{l} \ln\left(\frac{x+l}{x}\right)$$

Conservative Fields and Shielding

The Electrostatic Field Is Conservative

The integral of the electrostatic field around any closed path is zero:

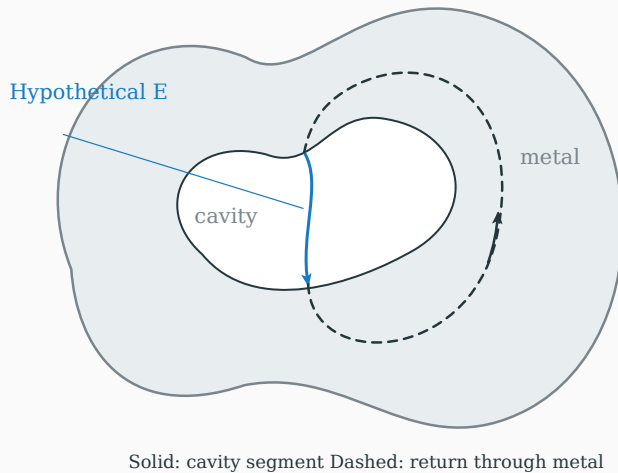
$$\oint \vec{E} \cdot d\vec{l} = 0$$

Consequences:

1. Electrostatic field lines cannot form closed loops. Along such a loop, the integral would be nonzero.
2. Field lines start on positive charges and end on negative charges, or extend to or from infinity.

An Empty Cavity in a Conductor

Within a closed, empty cavity inside a homogeneous conductor in electrostatic equilibrium, the electric field is exactly zero.



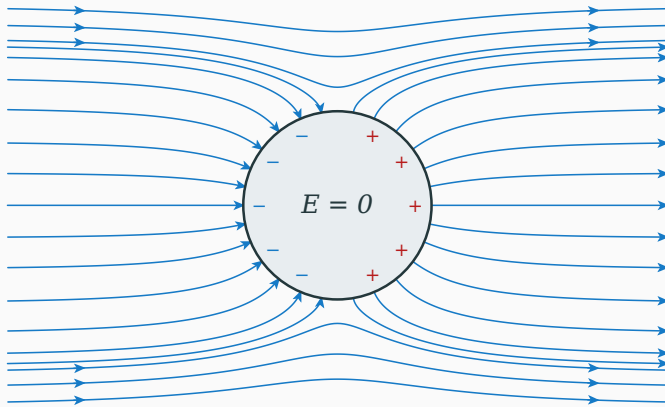
Suppose a field line exists inside the cavity. Complete a closed path through the conducting material.

The integral along the field line would be positive, while the part inside the conductor contributes zero. This contradicts

$$\oint \vec{E} \cdot d\vec{l} = 0.$$

Electrostatic Shielding

The absence of electrostatic fields in closed, charge-free conducting cavities is the basis of **Faraday cages**.



Neutral conducting sphere in a uniform external field

When a conductor is placed in an external electric field, free electrons redistribute on its surface so that the field inside the conducting material is zero.

The conductor may also carry an excess charge.

The Gradient of the Potential

From Potential to Electric Field

Why is V useful? We can obtain $\vec{E}$ by differentiation.

Consider (x, y, z) and $(x + \Delta x, y, z)$. The external work per unit charge is their potential difference:

$$\begin{aligned}\Delta V &= V(x + \Delta x, y, z) - V(x, y, z) \\ &\approx V(x, y, z) + \frac{\partial V}{\partial x} \Delta x - V(x, y, z) \\ &= \frac{\partial V}{\partial x} \Delta x.\end{aligned}$$

Here the expansion keeps terms to first order in Δx .

Components of the Electric Field

The same potential difference is

$$\Delta V = - \int \vec{E} \cdot d\vec{l} \approx -E_x \Delta x.$$

Comparing the expressions and taking $\Delta x \rightarrow 0$ gives

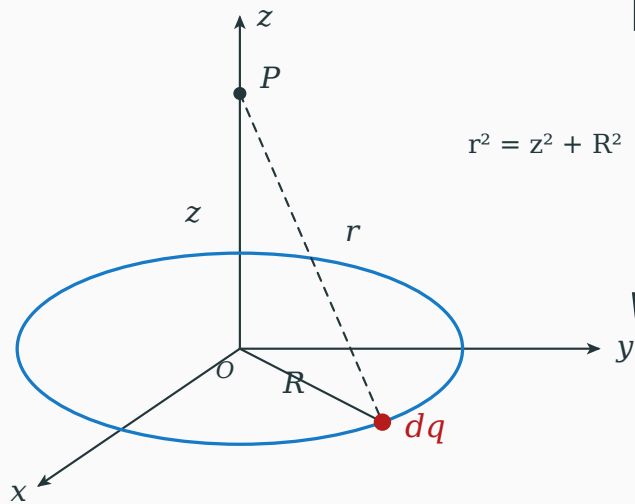
$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}.$$

$$\vec{E} = -\nabla V = -\frac{\partial V}{\partial x} \hat{x} - \frac{\partial V}{\partial y} \hat{y} - \frac{\partial V}{\partial z} \hat{z}$$

It is often easier to sum the scalar potentials of point charges and then differentiate than to sum their electric fields directly.

Example: Potential on the Axis of a Ring

Find the potential on the axis of a uniformly charged ring of radius R and total charge Q .



Each charge element contributes

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{dq}{\sqrt{z^2 + R^2}}.$$

With uniform density,

$$dq = \left(\frac{Q}{2\pi R} \right) R d\theta.$$

Problem 1.12 The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12),$$

where y is the distance (in miles) north, x the distance east of South Hadley.

- (a) Where is the top of the hill located?
- (b) How high is the hill?
- (c) How steep is the slope (in feet per mile) at a point 1 mile north and one mile east of South Hadley? In what direction is the slope steepest, at that point?

Ring Potential and Electric Field

Integrating around the ring gives

$$V(z) = \int_0^{2\pi} \frac{1}{4\pi\epsilon_0} \frac{Q}{2\pi R} \frac{R d\theta}{\sqrt{z^2 + R^2}}.$$

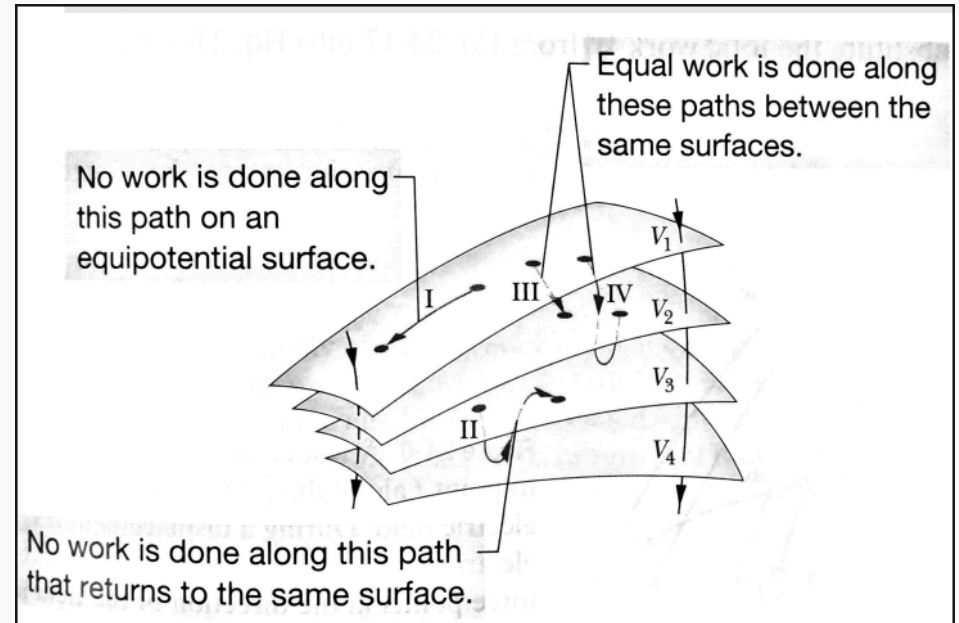
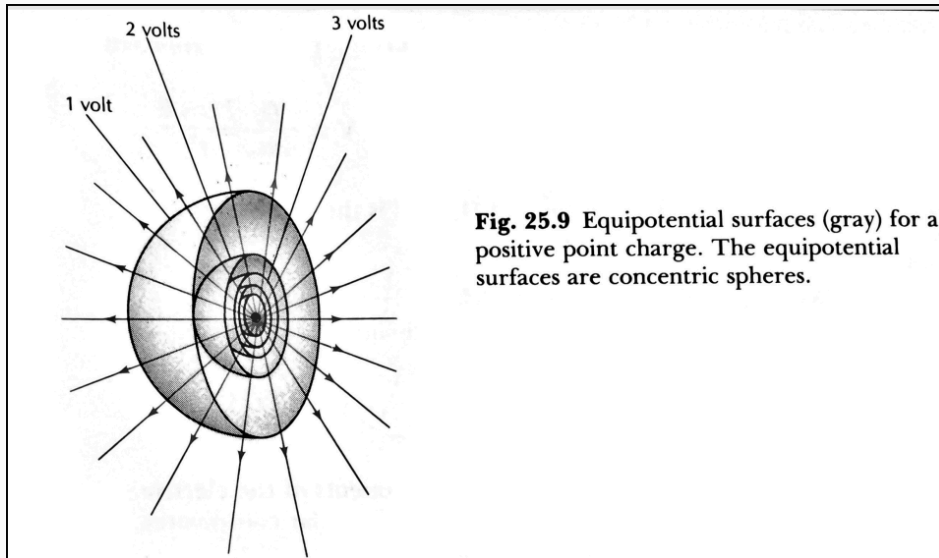
$$V(z) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{z^2 + R^2}}$$

Then

$$E_z = -\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon_0} \frac{Qz}{(z^2 + R^2)^{\frac{3}{2}}}.$$

Equipotential Surfaces

A mathematical surface on which the potential has a fixed, constant value is called an **equipotential surface**.



No work is done along an equipotential surface. Equal work per unit charge is required between the same two equipotential surfaces.

Electric Field and Equipotentials

The electric field is everywhere perpendicular to an equipotential surface wherever the field is nonzero.

This follows immediately from

$$dV = -\vec{E} \cdot d\vec{l}.$$

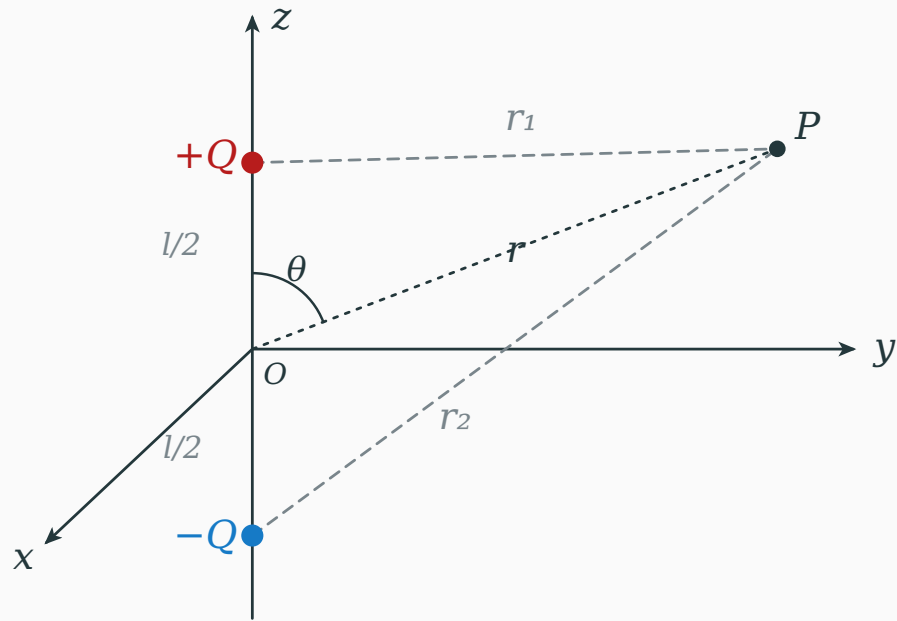
For a displacement tangent to an equipotential, $dV = 0$. Therefore,

$$\vec{E} \cdot d\vec{l} = 0.$$

Potential of a Dipole

Dipole Potential

Consider charges $+Q$ and $-Q$ separated by a distance l . The potential at P is the sum of their contributions:



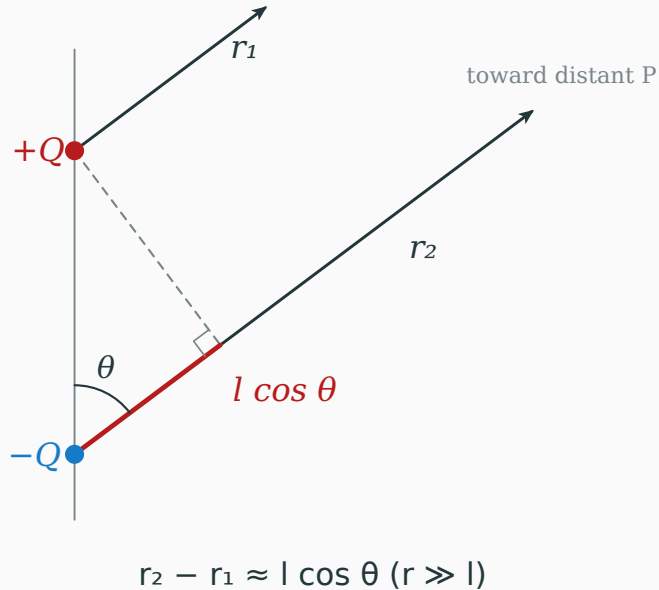
Thus,

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_1} - \frac{1}{4\pi\epsilon_0} \frac{Q}{r_2}.$$

$$V = \frac{Q}{4\pi\epsilon_0} \frac{r_2 - r_1}{r_1 r_2}.$$

The Far-Field Approximation

To obtain a simpler expression in terms of r and θ , assume $r \gg l$.



The lines from the two charges to P are nearly parallel:

$$r_1 \approx r_2 \approx r, \quad r_2 - r_1 \approx l \cos \theta.$$

With dipole moment magnitude $p = Ql$,

$$V \approx \frac{Q}{4\pi\epsilon_0} \frac{l \cos \theta}{r^2}.$$

$$V \approx \frac{p}{4\pi\epsilon_0} \frac{\cos \theta}{r^2}$$

Dipole Potential in Cartesian Coordinates

Express the result in rectangular coordinates:

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \cos \theta = \frac{z}{\sqrt{x^2 + y^2 + z^2}}.$$

Hence, for a dipole directed along $\hat{z}$,

$$V \approx \frac{p}{4\pi\epsilon_0} \frac{z}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

Dipole Electric-Field Components

Taking derivatives of the far-field potential,

$$E_x = -\frac{\partial V}{\partial x} \approx \frac{p}{4\pi\epsilon_0} \frac{3zx}{(x^2 + y^2 + z^2)^{\frac{5}{2}}},$$

$$E_y = -\frac{\partial V}{\partial y} \approx \frac{p}{4\pi\epsilon_0} \frac{3zy}{(x^2 + y^2 + z^2)^{\frac{5}{2}}},$$

$$E_z = -\frac{\partial V}{\partial z} \approx -\frac{p}{4\pi\epsilon_0} \left(\frac{1}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} - \frac{3z^2}{(x^2 + y^2 + z^2)^{\frac{5}{2}}} \right).$$

